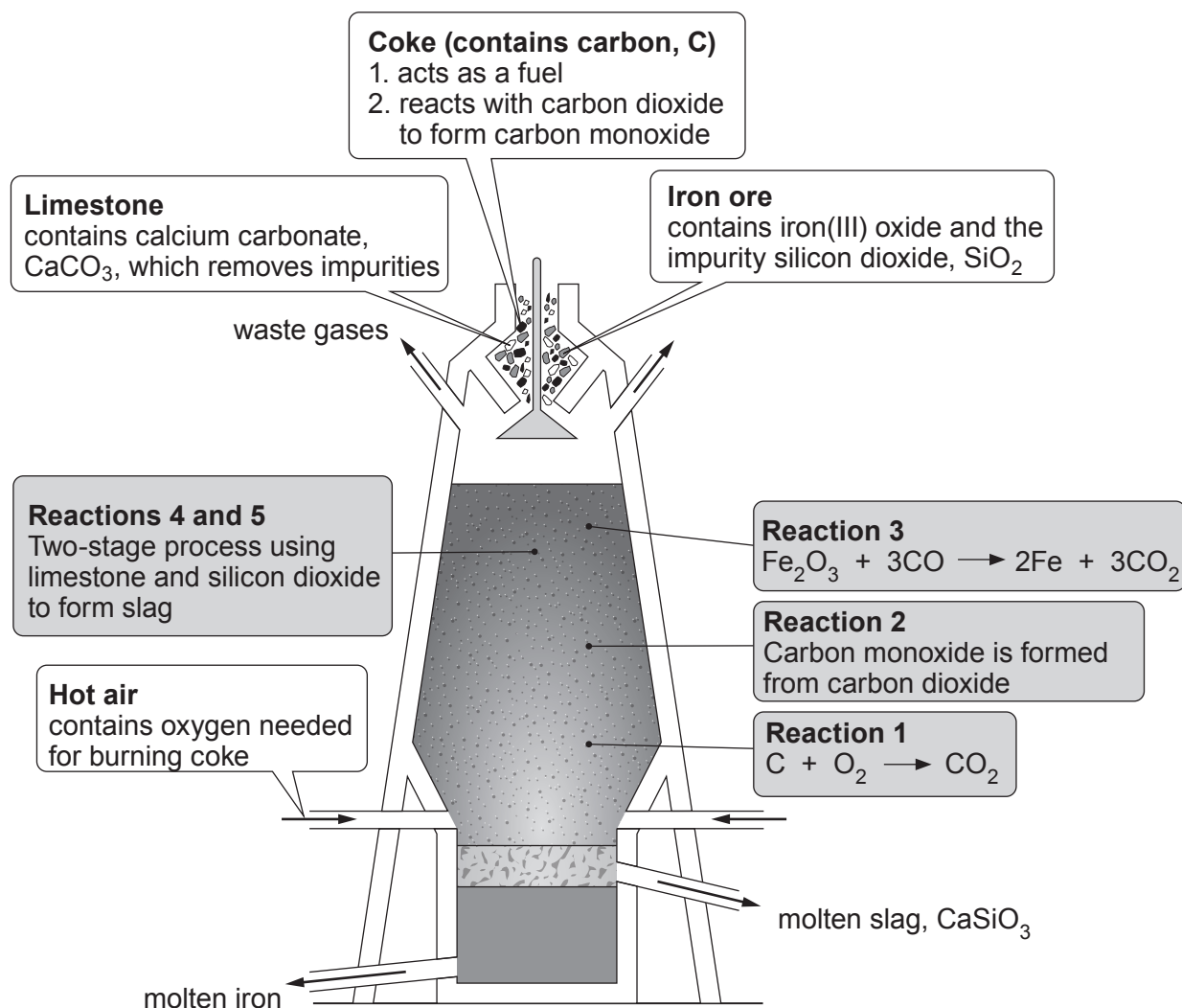


6. (a) Iron is extracted from its ore in the blast furnace.



Use information from the diagram and your knowledge to answer parts (i) and (ii).

- (i) Write a balanced symbol equation for reaction 2. [2]

.....

- (ii) Describe the two-stage process to form slag. [3]

.....

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.....

.....



(b) Iron(III) oxide reacts with dilute hydrochloric acid forming iron(III) chloride and water.

(i) Balance the symbol equation for this reaction. [1]



(ii) Sodium hydroxide solution can be used to detect the presence of aqueous iron(III) ions.

The symbol equation below represents the reaction occurring between solutions of sodium hydroxide and iron(III) chloride.

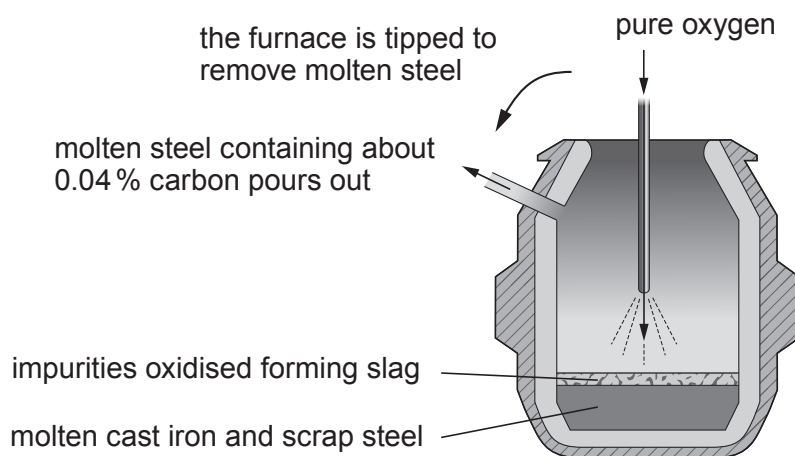


Write the **ionic** equation for the formation of the precipitate. [2]

.....



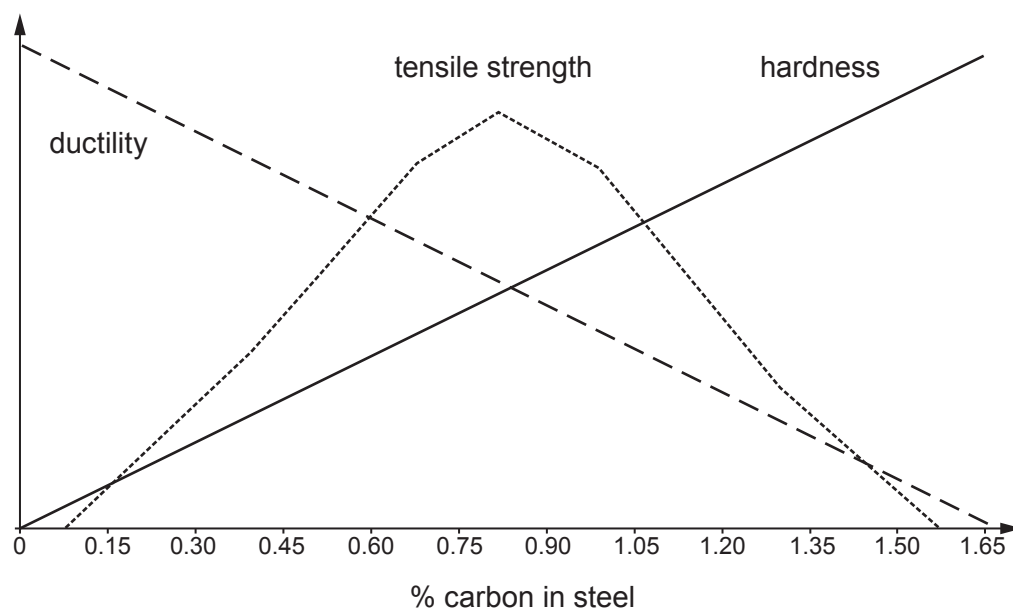
(c)

Steel manufacture in the UK

One of the major steelmaking processes used today in the UK is the Basic Oxygen Furnace, BOF. The raw materials for the BOF are cast iron from a blast furnace and scrap steel. Oxygen (>99.5% pure) is “blown” into the BOF at supersonic speed. The impurities are oxidised producing great quantities of heat which melts the scrap steel.

Basic Oxygen Furnace

Steel can be described in general terms as iron containing small amounts of carbon, to make it tougher and more ductile. There are many types of steel, each with its own specific chemical composition and properties to meet the needs of the many different applications. **Figure 1** below shows the relationships between the carbon content of steel and its ductility, tensile strength and hardness.

**Figure 1**

Ductility is a material's ability to be pulled into a wire.

Tensile strength is a measurement of the force required to pull something such as wire to the point where it breaks.

Hardness is a measure of how resistant a material is to permanent shape change when a compressive force is applied.



Figure 2 shows the percentage of carbon in various alloys of iron.

Name of alloy	Percentage of carbon
low carbon steel	0.0 – 0.6 %
medium carbon steel	0.6 – 0.8 %
high carbon steel	0.8 – 1.3 %
very high carbon steel	1.3 – 1.6 %

Figure 2

- (i) Tick (✓) the box next to the statement which best describes **one** way that production costs are reduced. [1]

high purity oxygen is used	☐
impurities are oxidised forming heat	☐
oxygen is blasted in at supersonic speed	☐
scrap steel is used in the process	☐

- (ii) Tick (✓) the box next to the statement which best describes the effect of increasing the percentage of carbon in steel from 0 % to 0.45 %. [1]

ductility increases, hardness increases	☐
tensile strength increases, ductility increases	☐
ductility decreases, tensile strength increases	☐
hardness increases, tensile strength decreases	☐

- (iii) A steel manufacturer wants to design an alloy with a high tensile strength but low ductility. Tick (✓) the box next to the approximate value for the percentage of carbon that should be included. [1]

0.2	☐	0.6	☐	1.0	☐	1.5	☐
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- (iv) Name the alloy which is the most easily pulled into a wire and withstands the least compressive force. [1]

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